

Elective course (Term 7)- Sept 2025



99.503: Nanoelectronics and Technology

Homework 3

(Submission due on Dec 02, 8PM)

1. Explain Extreme Ultraviolet (EUV) lithography procedure with the schematic of EUV system. Briefly discuss the challenges of using EUV lithography.
2. Describe how the performance of the capacitor affects the performance of the DRAM cell.
3. Briefly explain WRITE and READ operation of 6-T SRAM cell.
4. Explain the role of floating gate and tunnel oxide in flash memory cell. Describe the WRITE and READ operation in flash memory cell with the help of schematic.
5. Describe the necessity of ultra-high vacuum (UHV) conditions for nanoscale characterization techniques.
6. (a) Describe the key application of contact-AFM in characterizing the thin insulating materials.
(b) Explain the working of non-contact-AFM (Frequency modulation mode).
7. (a) Highlight the key applications of STM technique and challenges associated with STM characterization.
(b) Draw the force-distance graph in AFM and indicate the regions where contact and non-contact AFM works.
8. Explain the role of secondary electrons and backscattered electrons in characterization of materials using SEM.
9. Explain the physics behind the image (micrograph) formation when electron beam passes through sample in TEM.